

Lakshya JEE 2027

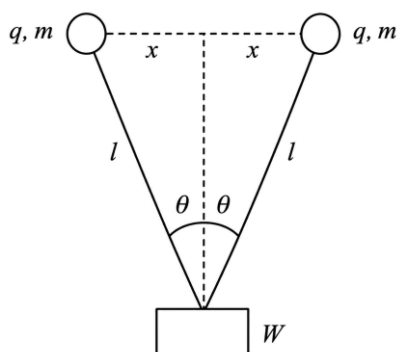
KPP - 02

Electrostatics

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1. Two identical He-filled spherical balloons each carrying a charge q are tied to a weight W with strings and float in equilibrium, as shown in figure.

- (a) Find the magnitude of q , assuming that the charge on each balloon acts as if it were concentrated at the center.
(b) Find the volume of each balloon. Take the density of He as ρ_{He} and the density of air as ρ_a . Ignore the weight of the unfilled balloons.



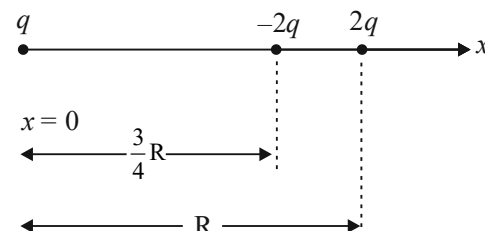
2. Two identical balls, each having a density ρ , are suspended from a common point by two insulating strings of equal length. Both the balls have equal mass and charge. In equilibrium, each string makes an angle θ with the vertical. Now, both the balls are immersed in a liquid. As a result, the angle θ does not change. The density of the liquid is σ . Find the dielectric constant of the liquid.
3. Two insulating small spheres are rubbed against each other and placed 1 cm apart. If they attract each other with a force of 0.1 N, how many electrons were transferred from one sphere to the other during rubbing?
4. Ten positively charged particles are kept fixed on the x -axis at points $x = 10$ cm, 20 cm, 30 cm, ..., 100 cm. The first particle has a charge 1.0×10^{-8} C, the second 8×10^{-8} C, the third 27×10^{-8} C and so on. The tenth particle has a charge 1000×10^{-8} C. Find the magnitude of the electric force acting on a 1 C charge placed at the origin.

5. Two charged particles having charge 2.0×10^{-8} C each are joined by an insulating string of length 1 m and the system is kept on a smooth horizontal table. Find the tension in the string.
6. Two identical pith balls are charged by rubbing against each other. They are suspended from a horizontal rod through two strings of length 20 cm each, the separation between the suspension points being 5 cm. In equilibrium, the separation between the balls is 3 cm. Find the mass of each ball and the tension in the strings. The charge on each ball has a magnitude 2.0×10^{-8} C. (take $g = 10$ m/s²).
7. Two small spheres, each having a mass of 20 g, are suspended from a common point by two insulating strings of length 40 cm each. The spheres are identically charged and the separation between the balls at equilibrium is found to be 4 cm. Find the charge on each sphere.
8. Two identical pith balls, each carrying a charge q , are suspended from a common point by two strings of equal length l . Find the mass of each ball if the angle between the strings is 2θ in equilibrium.
9. A particle having a charge of 2.0×10^{-4} C is placed directly below and at a separation of 10 cm from the bob of a simple pendulum at rest. The mass of the bob is 100 g. What charge should the bob be given so that the string becomes loose?
10. Two particles A and B having charges q and $2q$ respectively are placed on a smooth table with a separation d . A third particle C is to be clamped on the table in such a way that the particles A and B remain at rest on the table under electrical forces. What should be the charge on C and where should it be clamped?

11. Two identically charged particles are fastened to the two ends of a spring of spring constant 100 N m^{-1} and natural length 10 cm . The system rests on a smooth horizontal table. If the charge on each particle is $2.0 \times 10^{-8} \text{ C}$, find the extension in the length of the spring. Assume that the extension is small as compared to the natural length. Justify this assumption after you solve the problem.
12. A particle A having a charge of $2.0 \times 10^{-6} \text{ C}$ is held fixed on a horizontal table. A second charged particle of mass 80 g stays in equilibrium on the table at a distance of 10 cm from the first charge. The coefficient of friction between the table and this second particle is $\mu = 0.2$. Find the range within which the charge of this second particle may lie.
13. A particle A having a charge of $2.0 \times 10^{-6} \text{ C}$ and a mass of 100 g is placed at the bottom of a smooth inclined plane of inclination 30° . Where should another particle B, having same charge and mass, be placed on the incline so that it may remain in equilibrium?
14. Two identical balls, each having a charge of $2.00 \times 10^{-7} \text{ C}$ and a mass of 100 g , are suspended from a common point by two insulating strings each 50 cm long. The balls are held at a separation 5.0 cm apart and then released. Find
- the electric force on one of the charged balls
 - the components of the resultant force on it along and perpendicular to the string
 - the tension in the string
 - the acceleration of one of the balls.
- Answers are to be obtained only for the instant just after the release.
15. Two particles A and B, each carrying a charge Q , are held fixed with a separation d between them. A particle C having mass m and charge q is kept at the middle point of the line AB.
- If it is displaced through a distance x perpendicular to AB, what would be the electric force experienced by it.
 - Assuming $x \ll d$, show that this force is proportional to x .
 - Under what conditions will the particle C execute simple harmonic motion if it is released after such a small displacement?
- Find the time period of the oscillations if these conditions are satisfied.
16. Repeat the previous problem if the particle C is displaced through a distance x along the line AB.

17. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5 g/cc , the dielectric constant of water will be _____.
(Take density of water = 1 g/cc).

18. Three point charges q , $-2q$ and $2q$ are placed on x -axis at a distance $x = 0$, $x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $q = 2 \times 10^{-6} \text{ C}$ and $R = 2 \text{ cm}$, the magnitude of net force experienced by the charge $-2q$ is _____ N.



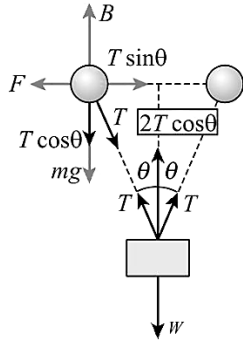
19. A point charge $q_1 = 4q_0$, is placed at origin. Another point charge $q_1 = -q_0$ is placed at $x = 12 \text{ cm}$. Charge of proton is q_0 . The proton is placed on x -axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.
20. Two identical positive charges Q each are fixed at a distance of ' $2a$ ' apart from each other. Another point charge q_0 with mass ' m ' is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 , will be:
- (1) $\sqrt{\frac{4\pi^3 \epsilon_0 m a^3}{q_0 Q}}$ (2) $\sqrt{\frac{q_0 Q}{4\pi^3 \epsilon_0 m a^3}}$
 - (3) $\sqrt{\frac{2\pi^2 \epsilon_0 m a^3}{q_0 Q}}$ (4) $\sqrt{\frac{8\pi^3 \epsilon_0 m a^3}{q_0 Q}}$
21. Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance d ($d \ll l$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result charges approach each other with a velocity v . Then as a function of distance x between them,
- (1) $v \propto x^{-1}$ (2) $v \propto x^{1/2}$
 - (3) $v \propto x$ (4) $v \propto x^{-1/2}$

Answer Key

1. (a) $q = \sqrt{8W \tan \theta \pi \epsilon_0 x^2}$, (b) $V = \frac{W}{2(\rho_a - \rho_{He})g}$
2. $\frac{\rho}{\rho - \sigma}$
3. 2×10^{11}
4. $4.95 \times 10^5 \text{ N}$
5. $3.6 \times 10^{-6} \text{ N}$
6. 8 g, $8 \times 10^{-2} \text{ N}$
7. $4.123 \times 10^{-8} \text{ C}$
8. $\frac{q^2 \cot \theta}{16\pi \epsilon_0 g l^2 \sin^2 \theta}$
9. $5.4 \times 10^{-9} \text{ C}$
10. $-(6 - 4\sqrt{2})q$, between q and $2q$ at a distance of $(\sqrt{2} - 1)d$ from q .
11. $3.6 \times 10^{-6} \text{ m}$
12. between $\pm 8.71 \times 10^{-8} \text{ C}$
13. 27 cm from the bottom
14. (a) 0.144 N, (b) zero, 0.095 N away from the other charge, (c) 0.986 N and (d) 0.95 m s^{-2} perpendicular to the string and going away from the other charge
15. (a) $\frac{Qqx}{2\pi \epsilon_0 \left(x^2 + \frac{d^2}{4}\right)^{3/2}}$, (c) $\left[\frac{m\pi^3 \epsilon_0 d^3}{Qq}\right]^{\frac{1}{2}}$
16. time period = $\left[\frac{\pi^3 \epsilon_0 m d^3}{2Qq}\right]^{\frac{1}{2}}$
17. (3)
18. (5440)
19. (24)
20. (1)
21. (4)

Solution

1. (a) $q = \sqrt{8W \tan \theta \pi \epsilon_0 x^2}$, (b) $V = \frac{W}{2(\rho_a - \rho_{He})g}$



- (a) $2T \cos \theta = W$, $T \sin \theta = F$ (Figure)

$$\text{or } \frac{\tan \theta}{2} = \frac{F}{W} \text{ or } F = W \frac{\tan \theta}{2}$$

$$\text{or } \frac{q^2}{4\pi\epsilon_0(2x)^2} = \frac{W \tan \theta}{2}$$

$$\text{or } q = \sqrt{8W \tan \theta \pi \epsilon_0 x^2}$$

- (b) $T \cos \theta + mg = B$

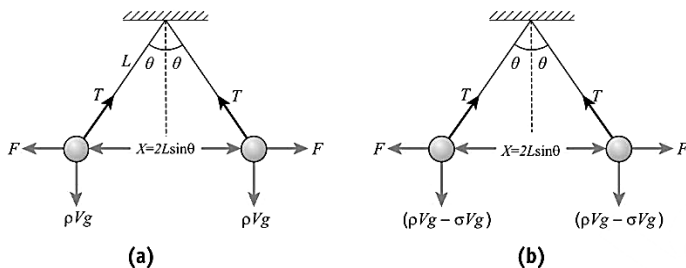
$$\Rightarrow \frac{W}{2} + V\rho_{He}g = V\rho_a g$$

$$\text{or } V = \frac{W}{2(\rho_a - \rho_{He})g}$$

2. $\frac{\rho}{\rho - \sigma}$

Let V be the volume of each ball, then the mass of each ball is $m = \rho V$. When the balls are in air, from the previous problem,

$$F = mg \tan \theta = \rho V g \tan \theta \quad \dots(i)$$



When the balls are suspended in liquid, the Coulombic force is reduced to $F' = F/K$ and apparent weight = weight - upthrust, i.e.,

$$W' = \rho V g - \sigma V g$$

According to the problem, angle θ is unchanged. Therefore,

$$F' = W' \tan \theta = (\rho V g - \sigma V g) \tan \theta \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{F}{F'} = K = \frac{\rho V g}{\rho V g - \sigma V g} = \frac{\rho}{\rho - \sigma}$$

3. 2×10^{11}

$$F = 0.1 \text{ N}$$

$r = 1 \text{ cm} = 10^{-2}$ (As they rubbed with each other. So the charge on each sphere are equal)

$$\text{So, } F = \frac{kq_1q_2}{r^2} \Rightarrow 0.1 = \frac{kq^2}{(10^{-2})^2}$$

$$\Rightarrow q^2 = \frac{0.1 \times 10^{-4}}{9 \times 10^9} \Rightarrow q^2 = \frac{1}{9} \times 10^{-14}$$

$$\Rightarrow q = \frac{1}{3} \times 10^{-7}$$

$$1.6 \times 10^{-19} \text{ c}$$

Carries by 1 electron

$$1 \text{ c carried by } \frac{1}{1.6 \times 10^{-19}} \times 0.33 \times 10^{-7}$$

$$= 0.208 \times 10^{12} = 2.08 \times 10^{11}$$

4. $4.95 \times 10^5 \text{ N}$

Electric force feeled by 1 c due to $1 \times 10^{-8} \text{ c}$.

$$F_1 = \frac{k \times 1 \times 10^{-8} \times 1}{(10 \times 10^{-2})^2} = k \times 10^{-6} \text{ N.} \quad \text{electric force}$$

feeled by 1 c due to $8 \times 10^{-8} \text{ c}$.

$$F_2 = \frac{k \times 8 \times 10^{-8} \times 1}{(23 \times 10^{-2})^2} = \frac{k \times 8 \times 10^{-8} \times 10^2}{9}$$

$$= \frac{28k \times 10^{-6}}{4} = 2k \times 10^{-6} \text{ N.}$$

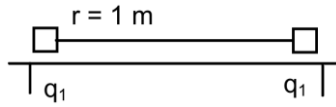
$$\text{Similarly } F_3 = \frac{k \times 27 \times 10^{-8} \times 1}{(30 \times 10^{-2})^2} = 3k \times 10^{-6} \text{ N}$$

So, $F = F_1 + F_2 + F_3 + \dots + F_{10} = k \times 10^{-6} (1 + 2 + 3 + \dots + 10) \text{ N}$

$$= k \times 10^{-6} \times \frac{10 \times 11}{2} = 55k \times 10^{-6}$$

$$= 55 \times 9 \times 10^9 \times 10^{-6} \text{ N} = 4.95 \times 10^3 \text{ N}$$

5. $3.6 \times 10^{-6} \text{ N}$



$$\text{Force exerted} = \frac{kq_1^2}{r^2}$$

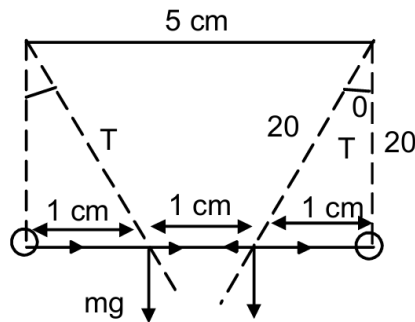
$$= \frac{9 \times 10^9 \times 2 \times 2 \times 10^{-16}}{1^2} = 3.6 \times 10^{-6} \text{ is the force exerted on the string}$$

6. 8.2 g, $8.2 \times 10^{-2} \text{ N}$



$q = 2.0 \times 10^{-8} \text{ C}$, $n = ?$, $T = ?$

$$\sin \theta = \frac{1}{20}$$



Force between the charges

$$F = \frac{Kq_1q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-8} \times 2 \times 10^{-8}}{(3 \times 10^{-2})^2} = 4 \times 10^{-3} \text{ N}$$

$$mg \sin \theta = F \Rightarrow m = \frac{F}{g \sin \theta} = \frac{4 \times 10^{-3}}{10 \times (1/20)}$$

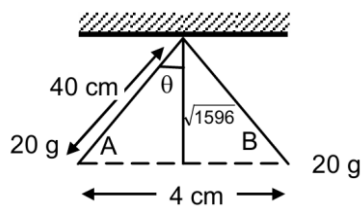
$$= 8 \times 10^{-3} = 8 \text{ gm}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{1}{400}} = \sqrt{\frac{400-1}{400}} = 0.99 \approx 1$$

So, $T = mg \cos \theta$

Or $T = 8 \times 10^{-3} \times 10 \times 1 = 8 \times 10^{-2} \text{ N}$

7. $4.123 \times 10^{-8} \text{ C}$



$T \cos \theta = mg \dots (1)$

$T \sin \theta = Fe \dots (2)$

Solving, (2)/(1) we get, $\tan \theta = \frac{Fe}{mg} = \frac{kq^2}{r} \times \frac{1}{mg}$

$$\Rightarrow \frac{2}{\sqrt{1596}} = \frac{9 \times 10^9 \times q^2}{(0.04)^2 \times 0.02 \times 9.8}$$

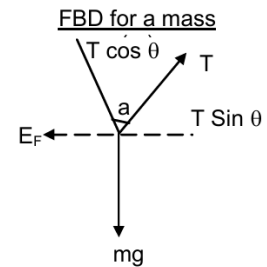
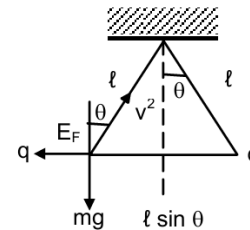
$$\Rightarrow q^2 = \frac{(0.04)^2 \times 0.02 \times 9.8 \times 2}{9 \times 10^9 \times \sqrt{1596}}$$

$$= \frac{6.27 \times 10^{-4}}{9 \times 10^9 \times 39.95} = 17 \times 10^{-16} \text{ C}^2$$

$$\Rightarrow q = \sqrt{17 \times 10^{-16}} = 4.123 \times 10^{-8} \text{ C}$$

8.

$$\frac{q^2 \cot \theta}{16\pi\epsilon_0 g l^2 \sin^2 \theta}$$



$$\text{Electric force} = \frac{kq^2}{(\ell \sin \theta + \ell \sin \theta)^2} = \frac{kq^2}{4\ell^2 \sin^2 \theta}$$

So, $T \cos \theta = mg$ (For equilibrium) $T \sin \theta = Ef$

$$\text{or } \tan \theta = \frac{Ef}{mg}$$

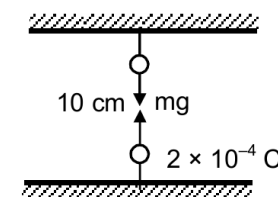
$$\Rightarrow mg = Ef \cot \theta = \frac{kq^2}{4\ell^2 \sin^2 \theta} \cot \theta$$

$$= \frac{q^2 \cot \theta}{\ell^2 \sin^2 \theta 16\pi\epsilon_0}$$

$$\text{or } m = \frac{q^2 \cot \theta}{16\pi\epsilon_0 \ell^2 \sin^2 \theta g} \text{ unit.}$$

9.

$5.4 \times 10^{-9} \text{ C}$



Mass of the bob = 100 g = 0.1 kg

So Tension in the string = $0.1 \times 9.8 = 0.98 \text{ N}$.

For the Tension to be 0, the charge below should repel the first bob.

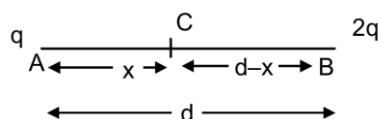
$$\Rightarrow F = \frac{kq_1q_2}{r^2}$$

$$T - mg + F = 0 \Rightarrow T = mg - f \quad T = mg$$

$$\Rightarrow 0.98 = \frac{9 \times 10^9 \times 2 \times 10^{-4} \times q_2}{(0.01)^2}$$

$$\Rightarrow q_2 = \frac{0.98 \times 1 \times 10^{-2} \times q_2}{9 \times 2 \times 10^5} = 5.4 \times 10^{-9} \text{ C}$$

10. $-(6 - 4\sqrt{2})q$, between q and $2q$ at a distance of $(\sqrt{2} - 1)d$ from q .



Let the charge on C = q

So, net force on c is equal to zero

$$\text{so } F_{AC} + F_{BC} = 0$$

$$\text{But } F_{AC} = F_{BC} \Rightarrow \frac{kqQ}{x^2} = \frac{k2qQ}{(d-x)^2}$$

$$\Rightarrow 2x^2 = (d-x)^2 \Rightarrow \sqrt{2}x = d-x$$

$$\Rightarrow x = \frac{d}{\sqrt{2}+1} = \frac{d}{(\sqrt{2}+1)} \times \frac{(\sqrt{2}-1)}{(\sqrt{2}-1)} = d(\sqrt{2}-1)$$

For the charge on rest, $F_{AC} + F_{AB} = 0$

$$(2.414)^2 \frac{kqQ}{d^2} + \frac{kq(2q)}{d^2} = 0$$

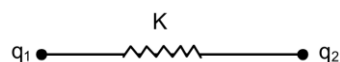
$$\Rightarrow \frac{kq}{d^2} [(2.414)^2 Q + 2q] = 0$$

$$\Rightarrow 2q = -(2.414)^2 Q$$

$$\Rightarrow Q = \frac{2}{-(\sqrt{2}+1)^2} q = -\left(\frac{2}{3+2\sqrt{2}}\right) q = -(0.343)q$$

$$= -(6 - 4\sqrt{2})q$$

11. $3.6 \times 10^{-6} \text{ m}$



$$K = 100 \text{ N/m}$$

$$l = 10 \text{ cm} = 10^{-1} \text{ m}$$

$$q = 2.0 \times 10^{-8} \text{ C Find } l = ?$$

Force between them

$$F = \frac{kq_1q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-8} \times 2 \times 10^{-8}}{10^{-2}}$$

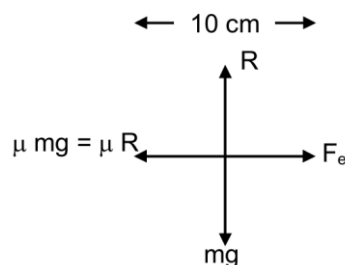
$$= 36 \times 10^{-5} \text{ N}$$

$$\text{So, } F = -kx$$

$$\text{or } x = \frac{F}{-K} = \frac{36 \times 10^{-5}}{100} = 36 \times 10^{-7} \text{ cm}$$

$$= 3.6 \times 10^{-6} \text{ m}$$

12. between $\pm 8.71 \times 10^{-8} \text{ C}$



$$q_A = 2 \times 10^{-6} \text{ C, } M_b = 80 \text{ g, } \mu = 0.2$$

Since B is at equilibrium, So, $F_e = \mu R$

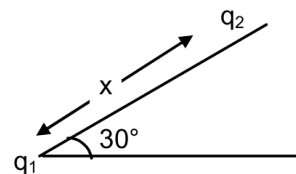
$$\Rightarrow \frac{Kq_Aq_B}{r^2} = \mu R = \mu m \times g$$

$$\Rightarrow \frac{9 \times 10^9 \times 2 \times 10^{-6} \times q_B}{0.01} = 0.2 \times 0.08 \times 9.8$$

$$\Rightarrow q_B = \frac{0.2 \times 0.08 \times 9.8 \times 0.01}{9 \times 10^9 \times 2 \times 10^{-6}} = 8.7 \times 10^{-8} \text{ C}$$

$$\text{Range} = \pm 8.7 \times 10^{-8} \text{ C}$$

13. 27 cm from the bottom



$$q_1 = 2 \times 10^{-6} \text{ C Let the distance be r unit}$$

$$\therefore F_{\text{repulsion}} = \frac{kq_1q_2}{r^2}$$

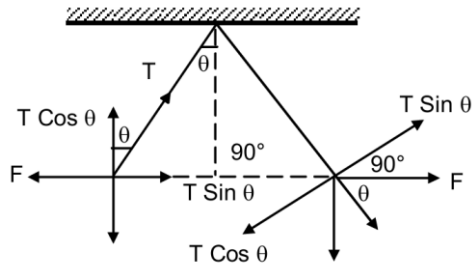
$$\text{For equilibrium } \frac{kq_1q_2}{r^2} = mg \sin \theta$$

$$\Rightarrow \frac{9 \times 10^9 \times 4 \times 10^{-12}}{r^2} = m \times 9.8 \times \frac{1}{2}$$

$$\Rightarrow r^2 = \frac{18 \times 4 \times 10^{-3}}{m \times 9.8} = \frac{72 \times 10^{-3}}{9.8 \times 10^{-1}} = 7.34 \times 10^{-2} \text{ metre}$$

$$\Rightarrow r = 2.70924 \times 10^{-1} \approx \text{metre } 27 \text{ cm from the bottom.}$$

14. (a) 0.144 N, (b) zero, 0.095 N away from the other charge, (c) 0.986 N and (d) 0.95 m s⁻² perpendicular to the string and going away from the other charge



$$q_1 = q_2 = 2 \times 10^{-7} \text{ C} \quad m = 100 \text{ g}$$

$$l = 50 \text{ cm} = 5 \times 10^{-2} \text{ m} \quad d = 5 \times 10^{-2} \text{ m}$$

(a) Now Electric force

$$F = K \frac{q^2}{r^2} = \frac{9 \times 10^9 \times 4 \times 10^{-14}}{25 \times 10^{-4}} \text{ N}$$

$$= 14.4 \times 10^{-2} \text{ N} = 0.144 \text{ N}$$

(b) The components of the resultant force along it is zero because mg balances $T \cos \theta$, as shown in the figure. So, $F = T \sin \theta$

$\perp$ to the string $F_{\text{net}} = T \cos \theta - mg \sin \theta$

$$= 0.144(1) - \frac{100}{1000} \times 10 \times \frac{1}{20}$$

$$= 0.144 - \frac{1}{20} = 0.094 \text{ N}$$

(c) Tension in the string,

$$T \sin \theta = F \quad \dots(1)$$

$$T \cos \theta = mg \quad \dots(2)$$

Squaring equations (1) and (2) and adding, we get

$$T^2 = F^2 + (mg)^2$$

$$= (0.144)^2 + (0.1 \times 9.8)^2$$

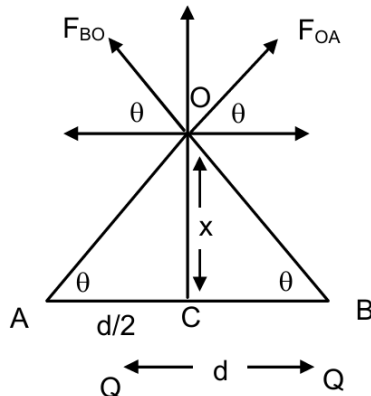
$$\Rightarrow T^2 = 0.986 \text{ N}$$

(d) The acceleration of one of the balls

$$a = \frac{F_{\text{net}}}{m} = \frac{0.094}{0.1}$$

$$= 0.94 \text{ m/s}^2.$$

15. (a) $\frac{Qqx}{2\pi\epsilon_0 \left(x^2 + \frac{d^2}{4}\right)^{3/2}}$, (c) $\left[\frac{m\pi^3\epsilon_0 d^3}{Qq}\right]^{1/2}$



(a) Let Q = charge on A and B Separated by distance d .

q = charge on c

displaced $\perp$ to $-AB$

$$\text{So, force on } 0 = \vec{F}_{AB} + \vec{F}_{BO}$$

$$\text{But } F_{AO} \cos \theta = F_{BO} \cos \theta$$

So, force on '0' in due to vertical component.

$$\vec{F} = F_{AO} \sin \theta + F_{BO} \sin \theta \quad |F_{AO}| = |F_{BO}|$$

$$= 2 \frac{KQq}{\left((d/2)^2 + x^2\right)} \sin \theta \quad F = \frac{2KQq}{\left((d/2)^2 + x^2\right)} \sin \theta$$

$$= \frac{4 \times 2 \times kQq}{\left(d^2 + 4x^2\right)} \times \frac{x}{\left[(d/2)^2 + x^2\right]^{1/2}}$$

$$= \frac{2kQq}{\left[(d/2)^2 + x^2\right]^{3/2}} x = \text{Electric force} \Rightarrow F \propto x$$

(b) When $x \ll d$

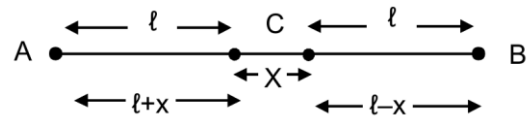
$$F = \frac{2kQq}{\left[(d/2)^2 + x^2\right]^{3/2}} x \quad x \ll d$$

$$\Rightarrow F = \frac{2kQq}{(d^2/4)^{3/2}} x \Rightarrow F \propto x$$

$$(c) a = \frac{F}{m} = \frac{1}{m} \left[\frac{2kQqx}{\left[(d^2/4) + \ell^2\right]} \right]$$

$$\text{So time period } T = 2\pi \sqrt{\frac{\ell}{g}} = 2\pi \sqrt{\frac{\ell}{a}}$$

16. time period = $\left[\frac{\pi^3 \epsilon_0 m d^3}{2Qq} \right]^{1/2}$



$$F_{AC} = \frac{KQq}{(\ell+x)^2} \quad F_{CA} = \frac{KQq}{(\ell-x)^2}$$

$$\text{Net force} = KQq \left[\frac{1}{(\ell-x)^2} - \frac{1}{(\ell+x)^2} \right]$$

$$= KQq \left[\frac{(\ell+x)^2 - (\ell-x)^2}{(\ell+x)^2 (\ell-x)^2} \right] = KQq \left[\frac{4\ell x}{(\ell^2 - x^2)^2} \right]$$

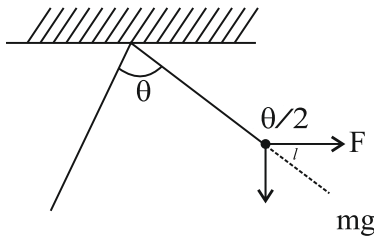
$x \ll \ell = d/2$ neglecting x w.r.t. ℓ We get

$$\text{net } F = \frac{KQq 4\ell x}{\ell^4} = \frac{KQq 4x}{\ell^3}$$

$$\text{acceleration} = \frac{4KQqx}{m\ell^3}$$

$$\begin{aligned}\text{Time period} &= 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}} \\ &= 2\pi \sqrt{\frac{x m \ell^3}{4 K Q q x}} = 2\pi \sqrt{\frac{m \ell^3}{4 K Q q}} \\ &= \sqrt{\frac{4 \pi^2 m \ell^3 4 \pi \epsilon_0}{4 Q q}} = \sqrt{\frac{4 \pi^3 m \ell^3 \epsilon_0}{Q q}} \\ &= \sqrt{4 \pi^3 m d^3 \epsilon_0 8 Q q} = \left[\frac{\pi^3 m d^3 \epsilon_0}{2 Q q} \right]^{1/2}\end{aligned}$$

17. (3)



If spheres are in air,

$$\tan \frac{\theta}{2} = \frac{F}{mg} = \frac{q^2}{4 \pi \epsilon_0 r^2 mg}$$

If spheres are in water,

$$\tan \frac{\theta}{2} = \frac{F'}{mg'} = \frac{q^2}{4 \pi k \epsilon_0 r^2 mg'}$$

Since angle remains same.

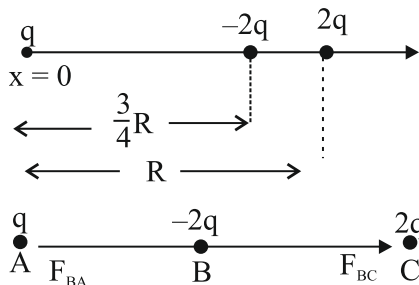
$$\Rightarrow \frac{q^2}{4 \pi \epsilon_0 r^2 mg} = \frac{q^2}{4 \pi k \epsilon_0 r^2 mg'}$$

$$\Rightarrow g = kg' = g \left(1 - \frac{1}{1.5} \right) k$$

$$\therefore k = 3$$

18. (5440)

From coulomb's law



$$F_{BA} = \frac{Kq(2q)}{\left(\frac{3}{4}R\right)^2} = \frac{32 Kq^2}{9R^2} \text{ and}$$

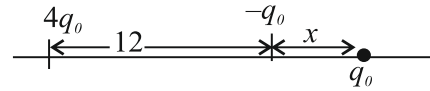
$$F_{BC} = \frac{K(2q)(2q)}{\left(\frac{R}{4}\right)^2} = \frac{64 Kq^2}{R^2}$$

$$F_B = F_{BC} - F_{BA} = \frac{544 Kq^2}{9R^2} B$$

$$= \frac{544 \times 9 \times 10^9 \times (2 \times 10^{-6})^2}{9 \times (2 \times 10^{-2})^2} = 5440 \text{ N}$$

19. (24)

Let x be the distance from charge $-q_0$ where electric force is zero.



Using the Coulomb's law

$$\frac{1}{4 \pi \epsilon_0} \cdot \frac{q_0^2}{x^2} = \frac{1}{4 \pi \epsilon_0} \cdot \frac{4q_0^2}{(x+12)^2}$$

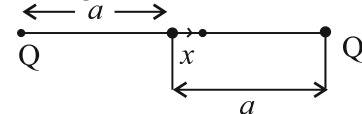
$$\Rightarrow x + 12 = 2x$$

$$x = 12$$

$$\text{Distance from origin} = x + 12 = 24 \text{ cm.}$$

20. (1)

Force on charge 'Q'



$$F = \frac{KQq_0}{(a-x)^2} - \frac{KQq_0}{(a+x)^2}$$

$$= KQq_0 \left(\frac{4ax}{(a^2 - x^2)^2} \right) \text{ as } a < x$$

$$\text{So, } F = \frac{KQq_0 \times 4ax}{a^4} = \frac{4Kxq_0Q}{a^3}$$

$$\text{As, } F = m\omega^2 x$$

$$\text{So, } m\omega^2 = \frac{4KQq_0}{a^3}$$

$$\omega = \sqrt{\frac{4KQq_0}{ma^3}} \Rightarrow T = \sqrt{\frac{4\pi^3 \epsilon_0 ma^3}{q_0 Q}}$$

21. (4)

From figure

$$T \cos \theta = mg$$

$$T \sin \theta = F_e$$

Dividing equation (ii) by (i), we get

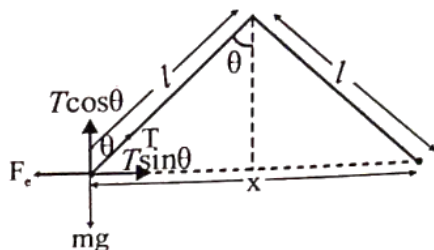


$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{F_e}{mg}$$

$$\Rightarrow F_e = mg \tan \theta$$

$$\Rightarrow \frac{kq^2}{x^2} = mg \tan \theta$$

$$\Rightarrow q^2 = \frac{x^2 mg \tan \theta}{k}$$



Since θ is small

$$\therefore \tan \theta \approx \sin \theta = \frac{x}{2l}$$

$$\therefore q^2 = \frac{x^3 mg}{2kl} \Rightarrow q^2 \propto x^{3/2}$$

$$\Rightarrow \frac{dq}{dt} \propto \frac{3}{2} \sqrt{x} \frac{dx}{dt} = \frac{3}{2} \sqrt{x} v$$

Since $\frac{dq}{dt} = \cos \omega t$, $\Rightarrow v \propto x^{-1/2}$



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>